An Anderson–Rubin Test of Sectoral-Composition Optimality at the Municipality Level

Specification, Identification, and the Reduced-Form Equivalence

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Abstract

This document is the technical reference for the Anderson–Rubin (AR) optimality test that is the inferential object of the paper. It (i) derives the directional-derivative condition that grounds the null $H_0 : \beta = \mathbf{0}$ in the envelope theorem; (ii) maps the shift-share identification framework of Borusyak, Hull, and Jaravel (2022) (BJS) onto our political-turnover instrument and discusses the levels-versus-changes choice formally; (iii) writes the instruments out from primitives, including the office-specific pre-election window and the two timing variants; (iv) shows that the AR test of $H_0 : \beta = \mathbf{0}$ is exactly the joint Wald test on the instruments in the reduced-form GDP equation, and that this equivalence is what makes the estimator of script 54 a weak-instrument-robust optimality test; (v) treats the volume-channel control formally; and (vi) collects the inference questions raised by spatial correlation in shift-share designs. The document is intentionally compact and presupposes the construction in `docs/methodology/regs_specification.tex`.

Setup and notation

We index municipalities by $m \in \{1, \dots, M\}$ ($M = 5{,}570$), sectors by $j \in \{1, \dots, J\}$, years by $t \in \{2002, \dots, 2017\}$, offices by $\ell \in \{\mathrm{M}, \mathrm{G}, \mathrm{P}\}$ (mayor, governor, president), and parties by p. Lower-case Latin letters f, g index firms.

The endogenous object is the vector of *sector employment shares*

$$s_{jmt} \;\equiv\; \frac{n_{jmt}}{n_{mt}}, \qquad n_{mt} \equiv \sum_{j=1}^{J} n_{jmt}, \qquad \sum_{j=1}^{J} s_{jmt} \;=\; 1,$$

constructed from RAIS firm–municipality–year employment counts. Stacked across j, $s_{mt} \equiv (s_{1mt}, \dots, s_{Jmt})' \in \Delta^{J-1}$ lies on the unit simplex; $\boldsymbol{\iota}' s_{mt} = 1$ and $\boldsymbol{\iota}' \Delta s_{mt} = 0$ for any change. The outcome is municipal log real GDP, $Y_{mt} \equiv \log(\mathrm{GDP}_{mt})$, and $\Delta Y_{mt} \equiv Y_{mt} - Y_{m,t-1}$.

We propose the following notation for BNDES sector credit shares to keep them distinct from employment shares:

$$s^{\mathrm{B}}_{jmt} \;\equiv\; \frac{\mathrm{BNDES}_{jmt}}{\sum_k \mathrm{BNDES}_{kmt}}, \tag{1}$$

where BNDES_{jmt} denotes BNDES disbursements to sector j in municipality m in year t. BNDES shares are not the estimand under the framing of D24; they are the upstream mechanism through which the political shock is transmitted to employment composition. They enter the analysis only as auxiliary first-stage objects.

The total-volume control is the unit-free ratio

$$\mathrm{Vol}_{mt} \;\equiv\; \frac{\sum_j \mathrm{BNDES}_{jmt}}{\mathrm{GDP}_{m,t_0}}, \tag{2}$$

with t_0 the initial period of the panel ($t_0 = 2002$).

1 The optimality test: theory

Environment. A municipal economy in (m,t) allocates fixed factors across J sectors with shares $s_{mt} \in \Delta^{J-1}$. Local output (or any local welfare proxy) is

$$Y_{mt} \;=\; Y(s_{mt};\, \theta_{mt})\,,$$

where θ_{mt} collects municipality-time fundamentals (productivity, factor prices, infrastructure) and $Y(\cdot;\theta)$ is twice continuously differentiable in s on the relative interior of Δ^{J-1}. Externalities through s_{mt} are admitted: the framework does not assume independence of the sectoral production technologies.

The directional-derivative condition. A feasible reallocation direction is any $v \in \mathbb{R}^J$ with $\boldsymbol{\iota}'v = 0$, so the perturbation $s_{mt} + \varepsilon v$ remains on the simplex for small ε. At an interior optimum $s^*_{mt} \in \text{int}(\Delta^{J-1})$ of $Y(\cdot;\theta_{mt})$, the first-order condition is

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} Y(s^*_{mt} + \varepsilon v)\bigg|_{\varepsilon=0} \;=\; \nabla_s Y(s^*_{mt}) \cdot v \;=\; 0 \qquad \text{for every } v \perp \boldsymbol{\iota}. \tag{3}$$

Equivalently, the gradient is parallel to the simplex normal, $\nabla_s Y \propto \boldsymbol{\iota}$, i.e. the social marginal returns to factors are equalized across sectors. This is the local-optimality benchmark.

Remark 1 (Envelope intuition). Equation (3) is the envelope condition for the planner's problem $\max_{s\in\Delta^{J-1}} Y(s;\theta)$: at an interior maximizer, infinitesimal feasible perturbations have zero first-order welfare effect (Samuelson, 1947). Rejection of (3) therefore identifies that the observed allocation is not at an interior maximum, and the sign pattern of the gradient prescribes the reallocation direction that would raise Y.

From the gradient to a regression. A first-order Taylor expansion of Y around the observed allocation at $t-1$ yields, for any direction Δs_{mt} with $\boldsymbol{\iota}'\Delta s_{mt} = 0$,

$$\Delta Y_{mt} \;=\; \nabla_s Y(s_{m,t-1};\theta_{m,t-1})'\,\Delta s_{mt} \;+\; \alpha_m + \delta_t + u_{mt} \;+\; O(\|\Delta s_{mt}\|^2), \tag{4}$$

where α_m and δ_t absorb time-invariant municipal and common year-level components of $\theta_{m,t-1}$, and u_{mt} is the residual. Define $\beta \equiv \nabla_s Y(s_{m,t-1};\theta_{m,t-1})$; under the local-optimality hypothesis (3) this gradient lies in the span of $\boldsymbol{\iota}$, so its projection onto the within-simplex subspace $\{v : \boldsymbol{\iota}'v = 0\}$ is the zero vector. The structural equation in levels of Y (with simplex-feasible s) is therefore

$$Y_{mt} \;=\; \alpha_m + \delta_t + \beta'\, s_{mt} + \lambda\,\text{Vol}_{mt} + \varepsilon_{mt}, \tag{5}$$

and the null hypothesis is

$$H_0: \quad \beta = \mathbf{0}. \tag{6}$$

The volume term Vol_{mt} enters because the same political-turnover shock that perturbs the sectoral mix also perturbs the aggregate level of credit; β identifies the composition channel only after the level channel has been partialed out. Section 6 develops this.

Remark 2 (Why a level regression rather than a difference). s_{mt} varies across m both within and across years, and Y_{mt} is itself a level. With municipality fixed effects α_m, equation (5) identifies β from the within-municipality time-variation of s_{mt}; with year fixed effects δ_t, common trends in Y are absorbed. The specification is equivalent up to projection to a first-difference equation in ΔY_{mt} and Δs_{mt} when α_m and δ_t are present; we keep the levels form because the data are stored as a long muni-year panel in which Vol_{mt} is naturally a level (initial-period denominator), and because the simplex constraint $\boldsymbol{\iota}'s_{mt} = 1$ is more transparent than $\boldsymbol{\iota}'\Delta s_{mt} = 0$.

Simplex constraint. Because $\boldsymbol{\iota}'s_{mt} = 1$ holds in every (m,t) cell, the regressor matrix of s_{mt} on a constant has reduced rank. Two operationally equivalent solutions:

1. Drop one sector j_0 (the largest-mean sector by convention, D6) and interpret each β_j as the effect of substituting one unit of employment share from j_0 to j.

2. Residualize s_{mt} on $\boldsymbol{\iota}$ (within (m,t) this is degenerate) or, equivalently, project onto the within-simplex subspace.

We adopt option (1) throughout. All elements of the estimated $\hat{\beta}$ should be read as differences relative to j_0, and the AR test on the remaining $J-1$ coefficients is invariant to the choice of j_0.

Remark 3 (Optional input–output adjustment). If the planner cares about gross output through the Leontief multiplier, one may replace s_{mt} with the IO-adjusted shares $\tilde{s}_{mt} \equiv B' s_{mt}$ where $B = (I-A)^{-1}$ for a national input–output matrix A. The local-optimality null becomes $\tilde{\beta} = \mathbf{0}$ in $Y_{mt} = \alpha_m + \delta_t + \tilde{\beta}' \tilde{s}_{mt} + \lambda \mathrm{Vol}_{mt} + \varepsilon_{mt}$. The instruments and the AR construction below carry over without modification because $\tilde{s}_{mt}$ is a known linear transformation of s_{mt}.

2 Identification strategy

2.1 The shift-share form of the instrument

Define the firm-level baseline owner-party exposure $\omega^{\ell}_{fp,t}$ and the sector-level exposure $w^{\ell}_{jmp,t}$ as in the companion methodology note `docs/methodology/regs_specification.tex` (§1). The sector-level levels instrument is

$$Z^{\ell}_{jmt} \equiv \sum_{p} w^{\ell}_{jmp,t}\, \mathrm{Align}^{\ell}_{mpt}, \tag{7}$$

and the changes variant is

$$\Delta Z^{\ell}_{jmt} \equiv \sum_{p} w^{\ell}_{jmp,t}\, \Delta\mathrm{Align}^{\ell}_{mpt}, \qquad \Delta\mathrm{Align}^{\ell}_{mpt} \equiv \mathrm{Align}^{\ell}_{mpt} - \mathrm{Align}^{\ell}_{mp,t-1}. \tag{8}$$

Z^{ℓ}_{jmt} is a within-region inner product of *shares* $w^{\ell}_{jmp,t}$ (sector-municipality exposure to party p, computed over the office-specific pre-election window) and *shifts* $\mathrm{Align}^{\ell}_{mpt}$ (whether p holds office ℓ in m). It is the sector-aggregated counterpart of the firm-level instrument $\mathrm{FA}^{\ell}_{fmt} = \sum_p \omega^{\ell}_{fp,t} \mathrm{Align}^{\ell}_{mpt}$ via the owner-count-weighted aggregation identity $Z^{\ell}_{jmt} = \sum_{f \in \mathcal{F}^{\mathrm{pre}}_{jme}} (L_{f,t}/N^{\ell}_{jm,t}) \mathrm{FA}^{\ell}_{fmt}$.

2.2 BJS conditions, mapped to our setting

Borusyak, Hull, and Jaravel (2022) treat shift-share IVs as IV estimators in a *shock-level* sample, where the unit of identification is a (shock) rather than an (m,t) cell. In our notation the "shocks" are office-party-municipality cells $k \equiv (\ell, p, m, t)$ with shifts $g_k \equiv \mathrm{Align}^{\ell}_{mpt}$ (or $\Delta\mathrm{Align}^{\ell}_{mpt}$) and exposures $w^{\ell}_{jmp,t}$. We restate their three primitive identification assumptions and map each to our setting.

Condition 1 (BJS-1: Quasi-random shock assignment)**.** Conditional on a vector of shock-level controls q_k, $\mathbb{E}[g_k \mid \{w^{\ell}_{jmp,t}\}_j,\, q_k] = \mu(q_k)$ is the same for shocks with different exposure profiles. Equivalently, in our panel: among (ℓ, p, m, t) cells with the same controls q_k, who is aligned is as good as randomly assigned with respect to municipal GDP residuals.

Condition 2 (BJS-2: Many uncorrelated shocks)**.** The number of effective shocks K_{eff} is large, and shocks are uncorrelated with each other after conditioning on q_k.

Condition 3 (BJS-3: Non-concentration of average exposure)**.** The Herfindahl of the average exposure weights across shocks is small, so no single shock drives the IV estimate.

What each buys. Condition 1 is the analogue of the exclusion restriction. It is what allows the instrument Z^{ℓ}_{jmt} to be exogenous in the structural GDP equation. Condition 2 provides the law-of-large-numbers justification that shock-level orthogonality of g_k to municipal residuals translates into unit-level (muni-year) orthogonality of Z^{ℓ}_{jmt} to ε_{mt}. Condition 3 ensures that the asymptotic distribution of the IV estimator is non-degenerate and not driven by a handful of large-exposure shocks.

What can violate Condition 1 in our setting. The political-alignment shock $\text{Align}^{\ell}_{mpt}$ is not literally random: parties win office because voters prefer them. Quasi-random assignment is plausible only after conditioning on a sufficient set q_k of shock-level controls. The relevant candidates for q_k are: (a) state-by-year fixed effects (so identification is within-state contests); (b) lagged municipal political controls (a party's pre-period vote share); (c) lagged GDP and employment trends (so a party that systematically wins in growing municipalities is not confounded with growth); and (d) sector-by-year fixed effects in the first stage (section 4) so that nationwide sectoral cycles are absorbed at the shock level. The empirical strategy includes muni and year fixed effects, exposure controls $\text{EC}^{\ell}_{jm,t}$, and (in robustness) a pre-period GDP trend; see section 7.

2.3 Levels versus changes: a formal sufficient condition

The same shift-share construction admits two timing variants of the shift: $\text{Align}^{\ell}_{mpt} \in \{0,1\}$ in levels, or $\Delta\text{Align}^{\ell}_{mpt} \in \{-1,0,+1\}$ in changes. They differ in how the identifying variation is distributed over time.

Proposition 1 (Sufficient conditions for the levels and changes instruments). *Suppose $Y_{mt} = \alpha_m + \delta_t + \beta' s_{mt} + \lambda \text{Vol}_{mt} + \varepsilon_{mt}$ with $\mathbb{E}[\varepsilon_{mt} \mid \alpha_m, \delta_t] = 0$, and let $\mathcal{H}_{m,t-1} \equiv \sigma(Y_{m,t'}, \text{Vol}_{m,t'}, s_{m,t'}; t' < t)$ be the pre-period information set.*

(i) The levels *instrument Z^{ℓ}_{jmt} satisfies the BJS quasi-random assignment condition Condition 1 if, for every cycle e_ℓ,*

$$\mathbb{E}\Big[\text{Align}^{\ell}_{mp,e_\ell+\tau} \,\big|\, \{w^{\ell}_{jmp,e_\ell}\}_j,\, \mathcal{H}_{m,e_\ell-1},\, q_k\Big] \;=\; \mu_\tau(q_k), \qquad \tau \in \{1,2,3,4\}, \tag{9}$$

i.e. alignment status is uncorrelated with prior municipal trajectories conditional on q_k, throughout the entire post-election cycle.

(ii) The changes *instrument ΔZ^{ℓ}_{jmt} satisfies Condition 1 if, for every inauguration year $t = e_\ell + 1$,*

$$\mathbb{E}\Big[\Delta\text{Align}^{\ell}_{mpt} \,\big|\, \{w^{\ell}_{jmp,t}\}_j,\, \mathcal{H}_{m,t-1},\, q_k\Big] \;=\; \mu(q_k), \tag{10}$$

i.e. the turnover at e_ℓ is uncorrelated with prior trajectories conditional on q_k.

Discussion. Equation (9) requires conditional mean-independence of alignment from pre-period outcomes *at every horizon* within the electoral term. Equation (10) requires it only *at the moment of turnover*. The latter is empirically weaker: turnover is closer to a discontinuity at the election date and need only be unrelated to the trajectory entering the election, not to the entire path that follows. In practice this means the changes specification carries a milder requirement on q_k, consistent with the standard discontinuity logic.

In the levels specification, $\text{Align}^{\ell}_{mpt}$ is non-zero for the entire ~ 4-year cycle, so the pre-period control set q_k must absorb anything that correlates persistently within a cycle with both alignment and outcomes (e.g. a slow-moving regional growth differential). Including a pre-period GDP trend, lagged $Y_{m,t-1}$, or muni-specific linear trends are natural strengthenings; the 2002-fixed baseline in $\omega^{\ell}_{fp,t}$ also reduces the dependence on contemporaneous alignment-cycle dynamics.

Many-shocks count under Condition 2. The effective number of shocks differs sharply between the two variants. With $M = 5{,}570$, four mayoral cycles ($e_\ell \in \{2004, 2008, 2012, 2016\}$) and $P \approx 30$ parties, the total *candidate* shock count is on the order of $M \times P \times 4 \approx 6.7 \times 10^5$ for the mayor tier alone. The levels variant uses each shock for ~ 4 post-election years, but those four year-replicates contribute redundant information for shock-level inference (the shock value is constant within a cycle). The changes variant uses each shock once at inauguration, so its effective shock count is the same but its panel-level non-zero cell count is smaller by a factor of ~ 4. Condition 2 is therefore satisfied at comparable levels by both variants in terms of *distinct shocks*, but the changes variant has less redundancy and a more transparent shock-level interpretation. By contrast, the levels variant has more non-zero panel cells and thus more degrees of freedom for the muni-year regression–at the cost of a stronger Condition 1 requirement ((9)).

Operational rule. Levels instruments are the primary specification, on grounds of statistical power and continuity with the existing pipeline. The changes instruments are the robustness specification. If the two reject in agreement, the substantive conclusion is robust to whichever of (9) or (10) actually holds in the data; if they disagree, the discrepancy isolates exactly the part of the variation that (9) is asking about beyond (10).

2.4 Exclusion restriction

The exclusion restriction proper is that the instrument Z^{ℓ}_{jmt} enters Y_{mt} only through the sectoral employment composition s_{mt}, after conditioning on the volume control Vol_{mt} and on the controls of section 7. Three non-composition channels are conceivable and are addressed in turn:

1. *Volume of BNDES credit.* Political alignment changes both the composition and the level of BNDES credit. The level effect is partialled out by Vol_{mt} in (5). The exclusion restriction asks that, conditional on Vol_{mt}, Z^{ℓ}_{jmt} does not drive Y_{mt} through a residual volume channel.
2. *Non-BNDES political channels.* A new mayor may also change procurement, transfers, or local public investment in ways that move GDP. Condition 1 accommodates this only if those channels are orthogonal to the *sector-specific* pattern of $w^{\ell}_{jmp,t}$. We test this in robustness as a placebo: if Z^{ℓ}_{jmt} predicts municipal transfers or procurement after conditioning, the exclusion restriction is weakened.
3. *Direct partisanship effects on private behaviour.* Even absent any policy change, owners aligned with the new ruling party may invest more in their own firms because of confidence shocks. This would deliver a reduced-form effect on s_{mt} via firm-level investment that is not mediated by BNDES credit, but it is still a *composition* channel and therefore part of the estimand by construction–it is not an exclusion violation. The estimand is the GDP effect of compositional shifts driven by the political shock, whether transmitted through BNDES credit or through partisan-firm investment.

3 Instrument construction

We summarize the construction; the primitives are derived in `regs_specification.tex`.

Pre-election window. For office ℓ and year t, let $e_\ell(t)$ be the most recent election year for office ℓ on or before t. The pre-election window is

$$T^{\ell}_t \;\equiv\; \big[e_\ell(t)-4,\; e_\ell(t)-1\big] \cap [2002, 2017]. \tag{11}$$

Within an electoral term T^{ℓ}_t is constant in t; at cycle boundaries, $e_\ell(t)$ advances and T^{ℓ}_t updates. The 2003 governor/president cycle (whose pre-window is 1998–2001, fully outside the data) is dropped.

Sector-municipality baseline exposure. Pooled-count, owner-count-weighted (D3, D8):

$$w^{\ell}_{jmp,t} \;\equiv\; \frac{1}{|T^{\ell}_t|} \sum_{s\in T^{\ell}_t} \frac{\sum_{f\in\mathcal{F}(j,m)} L_{f,p,s}}{\sum_{f\in\mathcal{F}(j,m)} L_{f,s}}, \qquad \sum_p w^{\ell}_{jmp,t} \le 1, \tag{12}$$

where $\mathcal{F}(j,m)$ is the firm support of cell (j,m), and $L_{f,p,s}$ and $L_{f,s}$ are firm-level affiliated and total owner counts. The residual $1-\sum_p w^{\ell}_{jmp,t}$ is the unaffiliated-owner share, which contributes zero to Z^{ℓ}_{jmt} but enters the exposure control $\mathrm{EC}^{\ell}_{jm,t} \equiv \sum_p w^{\ell}_{jmp,t}$ that absorbs total political connectedness of cell (j,m).

Sector instruments and muni-level stacking. Equations (7)–(8) define the sector-level instruments. To take them to the muni-year level for the AR test, we *stack* sector and office:

$$Z_{mt} \;\equiv\; \big(Z^{\ell}_{jmt}\big)_{\ell\in\{\mathrm{M,G,P}\},\, j=1,\dots,J} \;\in\; \mathbb{R}^K, \qquad K \;=\; 3\cdot J. \tag{13}$$

At the production margin `policy_block_active` $\times$ `S3` with $J = 12$ active bins, $K = 36$ for the full mayor-governor-president stack and $K = 12$ for the mayor-only specification (the primary in the strategy memo at docs/strategy/ar_test_strategy.md, §3). At the secondary margin cnae_section × S3 with $J = 51$ active bins, $K = 51$ or $K = 153$.

Remark 4 (Wide vs. aggregated muni-level instrument). An alternative aggregation, "stack-and-summarize," would collapse Z^ℓ_{jmt} to a single muni-year object via $Z^\ell_{mt} = \sum_j w_j Z^\ell_{jmt}$ for some weights w_j. This reduces K but imposes an aggregation that pre-loads which sectors matter; we treat it as a robustness check. The wide stacking (13) keeps the cross-sector pattern of the shock visible to the test and makes the "$\beta = \mathbf{0}$" null intelligible coefficient by coefficient.

4 First stage

The first stage relating sectoral employment shares to the instruments is estimated on the muni-sector-year panel (Panel A) at the chosen aggregation margin:

$$s_{jmt} \;=\; \sum_\ell \pi^\ell_j \, Z^\ell_{jmt} \;+\; \sum_\ell \phi^\ell_j \, \mathrm{EC}^\ell_{jm,t} \;+\; \gamma_{jm} + \alpha_{jt} \;+\; \eta_{jmt}, \tag{14}$$

with municipality×sector fixed effects γ_{jm} (~muni_id^cnae_section in fixest), sector×year fixed effects α_{jt} (~cnae_section^year), and standard errors two-way clustered by muni_id and sector. The exposure control $\mathrm{EC}^\ell_{jm,t}$ absorbs the total pre-period political connectedness of (j,m) across all parties and is the level-restricted analogue of Goldsmith-Pinkham, Sorkin, and Swift's share-level controls (Goldsmith-Pinkham, Sorkin, and Swift, 2020).

The first-stage F-statistic, computed via fixest :: wald(·, keep = "^Z_") on the full set of sector-tier instruments, is the diagnostic for *Condition* 3 and for F2 informativeness in our identification chain. Per D20 it is not a validity gate for the AR test: weak first stages widen the AR confidence set but do not bias inference.

Remark 5 (Stacked first stage for the AR test). The AR equivalence in section 5 does not require the first stage (14) to be estimated separately; it is internally embedded in the reduced-form Wald test. We document (14) because the F-statistic from (14) is the relevant power diagnostic, and because the pattern of $\hat{\pi}^\ell_j$ across sectors is the natural mechanism narrative.

5 The AR test: reduced form

5.1 Derivation

Step 1: Structural equation. At the muni-year level, restate (5) as

$$Y_{mt} \;=\; \alpha_m + \delta_t + \beta' s_{mt} \;+\; \lambda \, \mathrm{Vol}_{mt} \;+\; \varepsilon_{mt}. \tag{15}$$

Stack the K sector-tier instruments as in (13). Aggregate the sector-level first stage (14) to the muni-year level by reading $s_{mt} \in \mathbb{R}^J$ as a vector and writing the muni-year first stage in matrix form:

$$s_{mt} \;=\; \alpha^{(s)}_m + \delta^{(s)}_t \;+\; \Pi \, Z_{mt} \;+\; \Phi \, \mathrm{EC}_{mt} \;+\; \mu \, \mathrm{Vol}_{mt} \;+\; \nu_{mt}, \tag{16}$$

where $\Pi \in \mathbb{R}^{J \times K}$ is the matrix of first-stage coefficients collected from (14) after the sector-by-sector slopes have been expanded into the wide format aligned with (13), and $\mathrm{EC}_{mt} \equiv (\mathrm{EC}^\ell_{jm,t})_{\ell,j}$ is the stacked exposure-control vector.

Step 2: Reduced form by substitution. Substitute (16) into (15) and collect terms:

$$\begin{aligned} Y_{mt} &= \alpha_m + \delta_t + \beta' \Big(\alpha^{(s)}_m + \delta^{(s)}_t + \Pi \, Z_{mt} + \Phi \, \mathrm{EC}_{mt} + \mu \, \mathrm{Vol}_{mt} + \nu_{mt} \Big) + \lambda \, \mathrm{Vol}_{mt} + \varepsilon_{mt} \\ &= \tilde{\alpha}_m + \tilde{\delta}_t + \underbrace{(\Pi'\beta)'}_{\equiv \gamma' \in \mathbb{R}^K} \; Z_{mt} + \underbrace{(\Phi'\beta)'}_{\equiv \psi'} \, \mathrm{EC}_{mt} + \underbrace{(\beta'\mu + \lambda)}_{\equiv \tilde{\lambda}} \, \mathrm{Vol}_{mt} + \underbrace{\beta'\nu_{mt} + \varepsilon_{mt}}_{\equiv \eta_{mt}}, \end{aligned} \tag{17}$$

where $\tilde{\alpha}_m = \alpha_m + \beta' \alpha_m^{(s)}$ and $\tilde{\delta}_t = \delta_t + \beta' \delta_t^{(s)}$ are absorbed by the muni and year fixed effects. Equation (17) is the *reduced-form GDP equation.*

Step 3: H_0 in the reduced form. Under $H_0 : \beta = \mathbf{0}$,

$$\gamma \;=\; \Pi' \beta \;=\; \mathbf{0} \in \mathbb{R}^K, \qquad \psi \;=\; \Phi' \beta \;=\; \mathbf{0}, \qquad \tilde{\lambda} \;=\; \lambda. \tag{18}$$

Equation (18) is the testable content. It says that under H_0 all K instruments enter the reduced-form GDP equation with zero coefficient. This is true *regardless of* Π: the implication holds whether the first stage is strong, weak, or zero.

Step 4: The AR test as the joint Wald on Z_{mt}. Estimate

$$Y_{mt} \;=\; \tilde{\alpha}_m + \tilde{\delta}_t + \gamma' Z_{mt} + \psi' \,\mathrm{EC}_{mt} + \tilde{\lambda}\,\mathrm{Vol}_{mt} + \eta_{mt}, \tag{19}$$

and test

$$H_0^{\mathrm{AR}} : \quad \gamma = \mathbf{0} \quad \Longleftrightarrow \quad H_0 : \; \beta = \mathbf{0}. \tag{20}$$

The cluster-robust Wald statistic for $\gamma = \mathbf{0}$ in (19) is the AR test statistic. Operationally:

```
mod_rf <- feols(log_gdp_pc ~ Z + EC + Vol | muni_id + year,
                data = dt, vcov = ~muni_id)
ar_stat <- fixest::wald(mod_rf, keep = "^Z_")$stat
```

5.2 Equivalence with the canonical AR statistic

The canonical Anderson–Rubin statistic of Anderson and Rubin (1949) for $H_0 : \beta = \beta_0$ in the simultaneous equation $Y = X\beta + u$ with instruments Z is

$$\mathrm{AR}(\beta_0) \;=\; \frac{(Y - X\beta_0)' P_Z (Y - X\beta_0)/K}{(Y - X\beta_0)' M_Z (Y - X\beta_0)/(N-K)}, \tag{21}$$

with $P_Z = Z(Z'Z)^{-1}Z'$ and $M_Z = I - P_Z$ after projection of all variables on the fixed-effect space. Under correct specification and $\beta = \beta_0$, $\mathrm{AR}(\beta_0) \sim F_{K,\,N-K-n_{\mathrm{FE}}}$ exactly under homoskedasticity, and asymptotically χ^2_K/K under heteroskedasticity-robust forms (Andrews, Stock, and Sun, 2019; Kleibergen, 2002).

Lemma 1 (Reduced-form equivalence at $\beta_0 = \mathbf{0}$). *At $\beta_0 = \mathbf{0}$, the AR statistic* (21) *reduces to the F-statistic for $\gamma = \mathbf{0}$ in the reduced-form regression* (19).

Sketch. At $\beta_0 = \mathbf{0}$, $Y - X\beta_0 = Y$ and (21) becomes the ratio of the explained sum of squares of Y on Z (divided by K) to the residual sum of squares (divided by $N-K$), which is exactly the F-statistic for the joint significance of Z in $Y = \gamma' Z + (\mathrm{FE}) + \eta$. With heteroskedasticity-robust / cluster-robust variance, the corresponding Wald form replaces the homoskedastic F and is the cluster-robust AR statistic of Finlay and Magnusson (2009). □

What this buys us. Lemma 1 is what makes the AR optimality test cheap to implement: the practitioner only needs to estimate (19) and run a Wald test on the instruments. No first stage is needed for the test; no $\hat{\beta}$ needs to exist. The first stage (14)–(16) is informative for power but not for size.

5.3 Weak-instrument robustness, many instruments, and CLR

The AR test has correct size under $H_0 : \beta = \mathbf{0}$ for any value of Π, including $\Pi = 0$. This follows from the fact that under H_0 the reduced-form coefficient $\gamma = \Pi'\beta$ vanishes regardless of Π. The 2SLS Wald test is not size-correct when Π is small, because its critical values rely on a non-degenerate first-stage limit (Andrews, Stock, and Sun, 2019; Dufour, 1997).

For $K > 1$, the conditional likelihood ratio (CLR) test of Moreira (2003) dominates AR in power for many parameter configurations. We treat CLR as a complementary robustness statistic; it is not yet implemented in the project pipeline.

For $K \to \infty$ asymptotics (the secondary margin `cnae_section` $\times$ `S3` with $K \approx 50\text{–}150$), Mikusheva and Sun (2022) provide the ridge-regularised jackknifed AR test that is valid for $K/N \to c \in (0,1)$. This is the relevant inference for the many-instrument robustness specification.

6 Volume control

The volume term Vol_{mt} in (5) partitions the political shock's effect on Y_{mt} into a level (or volume) channel and a composition channel. Without it, the test of $H_0 : \beta = \mathbf{0}$ would conflate the two: turnover both reallocates credit across sectors and changes its aggregate amount, and (3) concerns only the former.

Construction.

$$\mathrm{Vol}_{mt} \equiv \frac{\sum_j \sum_{f \in \mathcal{F}(j,m)} \mathrm{BNDES}_{fmt}}{\mathrm{GDP}_{m,2002}} = \frac{\mathrm{BNDES\_total}_{mt}}{\mathrm{GDP}_{m,2002}}. \tag{22}$$

The numerator is current-period total municipal BNDES disbursements; the denominator is initial-period municipal GDP (2002 R$). The ratio is unit-free and time-comparable across municipalities. The choice of the 2002 denominator fixes the scaling at a pre-treatment value, so Vol_{mt} varies in t only through the numerator.

Remark 6 (Why a pre-treatment denominator). Using contemporaneous GDP_{mt} as the denominator would induce $\mathrm{cor}(\mathrm{Vol}_{mt}, Y_{mt}) \neq 0$ mechanically through the denominator, bias the partialling-out of the volume channel, and create a "bad-control" problem for the AR test. The 2002 denominator avoids this.

Should Vol_{mt} be instrumented? Vol_{mt} is itself endogenous: the same political shocks that move sector shares move the aggregate. The natural instrument for Vol_{mt} is a muni-level aggregation of Z^{ℓ}_{jmt}, weighted by the pre-period sector shares; this is the muni-level shift-share for the level channel. The four candidate operationalisations from the strategy memo (A10) are:

1. *Pure AR.* OLS for both s_{mt} and Vol_{mt} in the reduced form (i.e., neither is treated as the structural endogenous variable). The AR test is on Z_{mt}; Vol_{mt} is included as a pre-determined control.
2. *Partial IV* (**baseline**, per D24). Treat s_{mt} as endogenous and instrumented by Z_{mt}; treat Vol_{mt} as exogenous. The reduced-form AR equation (19) includes Vol_{mt} on the right-hand side and tests $\gamma = \mathbf{0}$ on Z_{mt}.
3. *Full IV.* Instrument both s_{mt} and Vol_{mt}. This requires a separate instrument for Vol_{mt} that is not collinear with Z_{mt} in finite sample; the obvious candidate is a sum or PCA of Z^{ℓ}_{jmt} across j, but that creates near-collinearity with Z_{mt} itself.
4. *Mixed.* OLS on s_{mt}, IV on Vol_{mt}. Symmetric to (3) and similarly suffers from near-collinearity.

The baseline (2) is the operative specification of the paper. Approaches (1), (3), (4) enter as robustness. The choice between (1) and (2) is consequential only if Vol_{mt} is a strong correlate of ε_{mt} *after* partialling out Z_{mt} and the FE; under the exclusion restriction this residual correlation is bounded by the strength of non-political confounders of total credit.

Remark 7 (The collinearity tradeoff). Approaches (3) and (4) deliver consistent estimators of both β and λ if a separate instrument for Vol_{mt} exists. The candidate $\sum_j Z^{\ell}_{jmt}$ collapses across sectors and discards the identifying variation in cross-sector tilts that is precisely what powers the AR test. Building a non-trivial second instrument requires new variation (e.g. a national fiscal-cycle shock that scales the BNDES envelope without shifting its sectoral allocation). We do not pursue this here.

7 Controls and fixed effects

We document the role of each control / fixed effect and the assumption it enables.

Municipality fixed effect α_m. Absorbs all time-invariant municipal characteristics: geography, fixed infrastructure stocks, time-invariant institutional quality, fixed sectoral-comparative-advantage components of θ_m. With α_m, identification of β uses only *within*-muni variation in s_{mt}. Required for Condition 1 to be plausible: a level correlation between alignment frequency and economic level across municipalities is allowed.

Year fixed effect δ_t. Absorbs national-level shocks affecting all municipalities in the same year: common monetary policy, national IPCA inflation, national electoral cycles affecting all jurisdictions simultaneously. Required so the identifying variation in $\text{Align}^{\ell}_{mpt}$ is the cross-muni dispersion in who-rules-where in a given year, not the national alternation between Lula, Dilma, and Temer.

Exposure control $\text{EC}^{\ell}_{jm,t}$. $\text{EC}^{\ell}_{jm,t} = \sum_p w^{\ell}_{jmp,t}$ measures total pre-period political connectedness of (j, m) at office ℓ. It is the level-restricted analogue of Goldsmith-Pinkham, Sorkin, and Swift's share-controls strategy: it absorbs any direct effect of the share component of the SSIV that is not mediated by the shift component. In the AR reduced form (19), $\text{EC}^{\ell}_{jm,t}$ is included alongside Z^{ℓ}_{jmt} to remove this share-channel confounding from the test on γ.

Pre-period GDP trend (robustness). A muni-specific linear trend in t, fitted on $Y_{m,t-1}$ and earlier, can be included to harden Condition 1 against persistent muni-level growth differentials that correlate with party alignment. This is the practical manifestation of the q_k in Condition 1. We treat it as robustness because it costs degrees of freedom (1 parameter per municipality) and because the muni-by-year FE design already absorbs everything cross-cycle.

State×year fixed effect (robustness). $\zeta_{s(m)t}$ where $s(m)$ is the state of m absorbs governor-level shocks common to all municipalities in a state. Including it removes the variation exploited by the governor-tier instruments Z^{G}_{jmt}, so it is incompatible with using all three tiers; it is included as a within-state robustness exercise that uses only mayor-tier instruments.

What is not included by design. We do not include lagged sector shares $s_{m,t-1}$ as controls. Doing so would absorb the pre-period mix and break the within-simplex projection: the gradient $\nabla_s Y$ is identified from changes *around* the pre-period mix, not from levels conditional on it. This is the standard "Nickell bias"-style argument adapted to the simplex.

8 Inference

Primary clustering. Standard errors in (19) are clustered by `muni_id`. The number of clusters ($M = 5{,}570$) is large relative to the parameter dimension and to the number of effective shocks per cycle, so cluster-robust inference is well-calibrated. Finlay and Magnusson (2009) establish that the cluster-robust Wald statistic for $\gamma = \mathbf{0}$ in (19) is the cluster-robust AR test, with size correctly controlled under quite general within-cluster dependence structures.

Two-way clustering. For the sector-level first stage (14) we cluster two-way by municipality and sector. For the muni-year reduced form (19) the natural second clustering dimension is harder: clustering by *shock* (ℓ, p, m, t) is collinear with the by-muni clustering, and clustering by year would yield only $T = 16$ clusters.

Spatial correlation. Adão, Kolesár, and Morales (2019) show that residuals in shift-share designs are correlated across units that share similar exposure structures (here, similar $w^{\ell}_{jmp,t}$ profiles). Cluster-robust SEs by `muni_id` do not fully internalize this correlation, and may understate uncertainty. Two remedies:

1. State-level clustering as a robustness check. The 27 Brazilian states are coarse and trigger few-clusters concerns (Carter, Schnepel, and Steigerwald, 2017); we use the wild cluster bootstrap in this case.
2. The Adão–Kolesár–Morales correction (Adão, Kolesár, and Morales, 2019), computed at the shock level. This is the right inference under the BJS framework but is not yet implemented in the pipeline.

Concern: state-level governor shock and within-state correlation. Within a state, all municipalities receive the same $\text{Align}^{\text{G}}_{mpt}$ in t. The SSIV variation across municipalities in Z^{G}_{jmt} then comes only from heterogeneity in $w^{\text{G}}_{jmp,t}$, the pre-period sector-municipality exposure. If residuals are correlated within a state through governor-level non-BNDES channels (state-level transfers, court appointments, public investment), inference on the governor-tier instruments is the most exposed to Adão, Kolesár, and Morales (2019)'s correction. Mayor-tier instruments are less exposed because $\text{Align}^{\text{M}}_{mpt}$ varies across municipalities. President-tier instruments are the most exposed because $\text{Align}^{\text{P}}_{mpt}$ is constant nationally in any given year and is absorbed by δ_t entirely; the within-tier identifying variation is purely through $w^{\text{P}}_{jmp,t}$. We therefore report tier-specific AR statistics alongside the joint test.

Effective sample size and time-series content. The levels instrument Z^{ℓ}_{jmt} is a step function in t that is constant within each electoral cycle. With muni and year FE, the variation exploited is the cross-municipality dispersion in instrument step-changes at cycle boundaries (mayor: 2005, 2009, 2013, 2017; governor/president: 2007, 2011, 2015). Each municipality contributes ~ 3–4 independent step-changes per tier; the cross-municipality average has $\sim M \times 3$ degrees of freedom for the mayor tier alone. This is the relevant denominator for power, not the panel-level $M \times T$.

9 Summary

The Anderson–Rubin optimality test of this paper is operationally a Wald test on the instruments in the reduced-form GDP regression (19). The substantive content is the directional-derivative condition (3) of section 1, which equates the local-optimality null to $\beta = \mathbf{0}$ in the structural equation (5). The reduced-form equivalence (Lemma 1) is what makes the test (i) weak-instrument robust, (ii) joint across K sector-tier instruments, and (iii) implementable without producing a $\hat{\beta}$ point estimate when the first stage is weak. The shift-share framework of Borusyak, Hull, and Jaravel (2022) provides the identification conditions; Proposition 1 states the formal sufficient conditions for the levels and changes timing variants and shows that the changes variant carries a milder requirement on pre-period dynamics. The volume control Vol_{mt} partials out the level channel of the political shock, leaving only the composition channel as the object of the test. Inference is cluster-robust by municipality, with state-level and Adão, Kolesár, and Morales (2019) corrections in robustness.

Appendix: Notation glossary

Symbol	Object
m, j, t, ℓ, p	municipality, sector, year, office, party
f	firm; $\mathcal{F}(j,m)$ firm support of cell (j,m)
s_{jmt}	sector j employment share in (m,t), $\sum_j s_{jmt} = 1$
s^{B}_{jmt}	sector j BNDES credit share in (m,t)
Y_{mt}	log real GDP in (m,t)
β	J-vector of GDP gradient w.r.t. employment shares
Vol_{mt}	$\text{BNDES\_total}_{mt}/\text{GDP}_{m,2002}$
T^{ℓ}_{t}	pre-election window for office ℓ at year t, eq.(11)
$L_{f,p,t}$	owners of firm f affiliated with party p at t
$\omega^{\ell}_{fp,t}$	firm-level baseline party exposure, pooled-count
$w^{\ell}_{jmp,t}$	sector-level baseline party exposure, eq.(12)
$\text{Align}^{\ell}_{mpt}$	$\mathbf{1}\{$party p holds office ℓ in m at $t\}$
$\Delta\text{Align}^{\ell}_{mpt}$	alignment turnover ($\neq 0$ at inauguration)
Z^{ℓ}_{jmt}	sector-tier levels instrument, eq.(7)
ΔZ^{ℓ}_{jmt}	sector-tier changes instrument, eq.(8)
Z_{mt}	stacked muni-year instrument vector, eq.(13), dim. K
$\text{EC}^{\ell}_{jm,t}$	sector-level exposure control, $\sum_p w^{\ell}_{jmp,t}$
Π	first-stage matrix in eq.(16)
$\gamma = \Pi'\beta$	reduced-form coefficient on Z_{mt} in eq.(19)
AR, F_{AR}	Anderson–Rubin statistic / F-form
$H_0 : \beta = \mathbf{0}$	local-optimality null

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